

Package ‘driftsel’

October 24, 2013

Type Package

Title Distinguishing drift and natural selection as causes of phenotypic differentiation

Version 2.1.2

Date 2013-10-24

Author Markku Karhunen & Otso Ovaskainen

Maintainer Markku Karhunen <markku.karhunen@helsinki.fi>

Depends R (>= 2.12.0), MCMCpack, SparseM, corpcor

Description Neutrality test for quantitative traits

License GPL-2

LazyLoad yes

R topics documented:

driftsel	2
H.test	3
MH	4
neut.test	6
S.test	6
specimen	7
specimen2	8
threespines	9
viz.theta	10
viz.theta.exact	11
viz.traits	12
Index	13

driftsel

*Drift and selection in natural populations***Description**

This package implements the neutrality test described in the Genetics paper of Ovaskainen et al. (2011), and explained in more detail by Karhunen et al. (2013). Shortly, it fits a Bayesian animal model and compares it with a measure of neutral genetic differentiation. (This measure can be obtained by using the R package RAFM.) Alternatively, driftsel can also be used to estimate G matrices and effects of covariates on phenotypes in a panmictic population. The package includes the estimation method MH, the neutrality test, visualization aids and example datasets. Versions 2.0+ involve binary traits and the so-called H test. Versions 2.1+ improve mixing for G matrix estimation.

New features of version 2.1.2: viz.theta.exact added and viz.traits improved. Source code prettier.

Details

Package:	driftsel
Type:	Package
Version:	2.1.2
Date:	2013-10-24
License:	GPL-2
LazyLoad:	yes

Use package RAFM to analyze the neutral DNA. Subsequently, use driftsel to analyze the breeding design and to do the neutrality test. Note that the estimation function MH gives much information besides the neutrality test, e.g. the effects of fixed and random covariates on the traits of interest.

Author(s)

Markku Karhunen, <markku.karhunen@helsinki.fi>

References

Ovaskainen et al. (2011): A new method to uncover signatures of divergent and stabilizing selection in quantitative traits. *Genetics* 189: 621-632.

Karhunen et al. (2013): driftsel: an R package for detecting signals of natural selection in quantitative traits. *Molecular Ecology Resources* 13: 746-754.

Examples

```
data(specimen)
A <- specimen
samp <- MH(A$poster, A$ped, A$covars, A$traits, 12, 4, 1)
neut.test(samp$pop.ef, samp$G, samp$theta, silent=TRUE)
```

H.test

*Neutrality test for the posteriors obtained from MH***Description**

This function calculates the test statistic H (Karhunen et al. 2013b).

Usage

```
H.test(popefpost, Gpost, THpost, env, silent = T)
```

Arguments

popefpost	posterior of population effects from function MH
Gpost	posterior of G matrix
THpost	posterior of theta
env	matrix of environmental data, rows=subpopulations, columns=variables
silent	plot convergence checks?

Details

H.test works in a way much analogous to S.test. S.test asks, whether the population means are too near or far to the ancestral mean. H.test asks, whether the population means correlate with the environmental data more than would be expected on basis of shared evolutionary history. E.g. all polar bears are similar and different to brown bears, both because of shared climate and shared phylogeny. Does shared phylongey explain this all, or are the similarities suspiciously similar to similarities in let's say latitude?

Large values of H imply that the populations are more adapted to their environment than would be expected at random, i.e. on basis of random drift. E.g. $H > 0.95$ implies adaptation to the covariates in env with p value 0.05. (Small values of H are difficult to interpret; probably they mean that the covariates in env are just irrelevant regarding evolutionary adaptation.)

Value

The test statistic H, defined as the posterior probability that the observed product moment of environmental and phenotypic distance is greater than a similar product moment produced under the neutral model, i.e. in absence of natural selection.

References

Karhunen et al. (2013b): Evolution of quantitative traits by natural selection: a case study of nine-spined sticklebacks. *Evolution XX*: pp-pp.

Examples

```
# Analayzing specimen data by MCMC:
data(specimen)
A <- specimen
samp <- MH(A$poster, A$ped, A$covars, A$traits, 21, 7, 2) # a small posterior
```

```
# Let's make some environmental data!
dim(A$poster) # how many populations?
pops = 1:2
rain = runif(2, 0, 1000)
temp = runif(2, 0, 25)
wind = runif(2, 0, 360)
env = cbind(pops, rain, temp, wind)

# Let's calculate the test:
H <- H.test(samp$pop.ef, samp$G, samp$theta, env, silent=TRUE)
```

MH

A Metropolis-Hastings algorithm for quantitative genetics

Description

This function fits the animal model described by Ovaskainen et al. while sampling from the posterior of population-population coancestry coefficients.

Usage

```
MH(poster, ped, covars, traits, nmc, burnin, thin,
   blocks = NA, priors = NA, tmp = NA, binary = NA, alt = F)
```

Arguments

poster	posterior of AFM, a 3-dimensional array, see Details
ped	pedigree matrix (NOTE that these have to be R matrices, not e.g. data frames)
covars	covariates modelled as fixed effects, matrix
traits	traits of interest, matrix
nmc	total number of MC iterations
burnin	length of transient period
thin	thinning interval
blocks	covariates modelled as random effects, optional
priors	list of prior hyper-parameters, optional
tmp	tmpmax for sparse Cholesky decomposition, optional
binary	which traits are binary?
alt	is pedigree in alternative short format (as in specimen2)?

Details

Use the R package RAFM to analyze molecular markers, calculate posterior of theta, and use it as argument poster in this function. Use MH to analyze the phenotypic data. Please see specimen and specimen2 for formatting your data for use with this package. The observations have to be in the same order for every argument of MH. Blocks do not need to be coded as dummies (whereas categorical variables in covars need to be). Note that the arguments of MH are matrices or arrays.

If you do not wish to give any covariates, assign covars <- traits[,1] (which is ID). In this case, each trait is modelled by a constant term. In fact, a constant term is always used regardless of

the covariates, corresponding to `traits~covars+1` in standard regression models. It is the first 'fixed effect' in the output.

If you have binary data, code them as 0 and 1, and mention these traits in binary. If `binary=NA`, all data are assumed normally distributed within populations.

As regarding the priors, format the list according to `specimen$priors`. You can also use `priors$fixed[[2]]=NA`, unlike in `specimen`. This produces the model described by Ovaskainen et al., whereas typing a covariance matrix removes the link between the population means and the ancestral G matrix, potentially confusing the neutrality test. Use `priors=NA`, if you don't know what to do. The default priors are quite uninformative.

Try different values of argument `tmp`, if the program crashes on your data with 'Error: ... increase `tmpmax`'. High values of `tmp` should decrease the degree of sparseness in Cholesky decomposition, and allow you to run your data at the cost of increased computation time. If this error doesn't occur, don't touch `tmp`.

Regarding the output, posteriors of all parameters are given as three-dimensional arrays, last dimension referring to iteration. Output includes fixed effects, population means, ancestral G matrix, E matrix and theta matrix. Theta is the population-level coancestry matrix. Its posterior sample is a subset of the values given in poster, filtered according to their likelihood in the phenotypic data.

Remember to check the convergence of the Markov chains in the posterior sample of parameters of interest. Increase `nmc`, `burnin` and/or `thin`, if convergence is poor.

Value

<code>fixed.ef</code>	Posterior of fixed effects, dimension: covariate x trait x iteration
<code>pop.ef</code>	Posterior of subpopulation effects, dimension: population x trait x iteration
<code>G</code>	Posterior of G matrix, dimension: trait x trait x iteration
<code>E</code>	Posterior of E matrix, dimension: trait x trait x iteration
<code>theta</code>	Posterior of theta matrix, dimension: population x population x iteration

References

Ovaskainen et al. (2011): A new method to uncover signatures of divergent and stabilizing selection in quantitative traits. *Genetics* 189: 621-632.

Examples

```
# Basic usage:
data(specimen)
A <- specimen
samp <- MH(A$poster, A$ped, A$covars, A$traits, 2, 1, 1) # uninformative priors

# Using informative priors:
samp <- MH(A$poster, A$ped, A$covars, A$traits, 2, 1, 1, priors=A$priors)

# Modelling study-design blocks as random effects:
samp <- MH(A$poster, A$ped, A$covars, A$traits, 2, 1, 1, blocks=A$blocks)
names(samp)
mains <- samp$main.result # DO THIS, IF YOU USE RANDOM EFFECTS
mains$G
plot(mains$pop.ef[1,1,], type='l', xlab="iter", ylab="", main="") # one Markov chain
plot(mains$G[1,2,], type='l', xlab="iter", ylab="", main="") # another Markov chain
```

```
# Panmictic population:
nostruc <- 0*A$poster
samp <- MH(nostruc, A$ped, A$covars, A$traits, 2, 1, 1)

# Coding trait 2 as binary and re-running the analysis:
A$traits[,3] = 1*(A$traits[,3]>0) # e.g. positive attitude
samp <- MH(A$poster, A$ped, A$covars, A$traits, 2, 1, 1, binary=2)
```

neut.test	<i>Neutrality test for backward compatibility</i>
-----------	---

Description

neut.test() = S.test()

Value

Gives out exactly the same as S.test.

S.test	<i>Neutrality test for the posteriors obtained from MH</i>
--------	--

Description

This function calculates the test statistic S (see Ovaskainen et al. 2011 and Karhunen et al. 2013).

Usage

```
S.test(popefpost, Gpost, THpost, silent = T, G.off = F,
th.off = F, main = NA)
```

Arguments

popefpost	posterior of population effects from function MH
Gpost	posterior of G matrix
THpost	posterior of theta
silent	plot the trace of the test statistic?
G.off	ignore genetic correlations?
th.off	ignore signatures of gene flow?
main	title for the diagnostic plot, optional

Details

popefpost, Gpost and THpost should be \$pop.ef, \$G and \$theta from the output of MH. You can use silent=FALSE to investigate convergence in the neutrality test. Note that you don't necessarily want to see white noise on [0,1], but trendless noise in some subset of [0,1].

Value

The test statistic S . It measures how far the multivariate population means have drifted from the ancestral mean. $S=0.5$ implies perfect match with neutrality. $S=0$ and $S=1$ imply perfect match with stabilizing and diversifying selection, respectively. The test is known to be conservative, so that $S=0.99$ implies diversifying selection at least on the significance level of 0.99.

References

Ovaskainen et al. (2011): A new method to uncover signatures of divergent and stabilizing selection in quantitative traits. *Genetics* 189: 621-632.

Karhunen et al. (2013): driftsel: an R package for detecting signals of natural selection in quantitative traits. *Molecular Ecology Resources* XX: pp-pp.

Examples

```
data(specimen)
A <- specimen
samp <- MH(A$poster, A$ped, A$covars, A$traits, 21, 7, 2) # a small posterior
S <- S.test(samp$pop.ef, samp$G, samp$theta) # neutrality test
S <- S.test(samp$pop.ef, samp$G, samp$theta, G.off=TRUE) # does the value change?
```

specimen

Sample input for MH, data in long format

Description

Non-informative data that will roughly return the prior.

Usage

```
data(specimen)
```

Format

List of six:

poster RAFM output, dimension: population x population x iteration

ped Pedigree matrix

covars Covariate matrix

traits Traits of interest, matrix

blocks Study design blocks, matrix

prior List of prior hyper-parameters

Details

The component `ped` is the pedigree matrix. Individuals who have a number in the fourth column are interpreted as founders caught from the wild. Other individuals are interpreted as offspring (P1, P2 etc.). An individual should always appear before its offspring. The order of individuals should be the same in all data matrices (`ped`, `covars`, `traits` and `blocks`). IDs refer directly to sire and dam codes in `ped`. Selfing is allowed, so use separate codes for sires and dams, if it's impossible for your data.

The component `prior` consists of lists. First component is the mean and variance of the fixed and population effects. You can replace the covariance matrix by NA for a theoretically justified model. The components `$G` and `$E` give the degrees of freedom and the inverse scale matrix of the respective priors.

Component `$random` is a list such that one component is a similar parameter list as is used for `G` and `E` matrices. In this case, `length(specimen$prior$random)` is 1, because there is only one random effect, that of design blocks. If you use more columns (and custom priors), you need a longer list. If you use your own priors, it is your responsibility to check that they are conformable with the study design.

Examples

```
# How it looks:
data(specimen)
head(specimen$ped)
head(specimen$traits)
n = nrow(specimen$ped)

# Including another 'block effect':
specimen$blocks <- cbind(specimen$blocks, sample(1:2, n, replace=TRUE))

# Correcting prior accordingly:
specimen$priors$random <- list(specimen$priors$random, specimen$priors$random)
specimen$priors$random

# Default model:
specimen$priors$fixed[[2]] = NA
```

specimen2

Another sample input for MH, data in short format

Description

Short format for one-generation study designs.

Usage

```
data(specimen2)
```

Format

List of six:

poster RAFM output, dimension: population x population x iteration

ped Pedigree matrix, sires and dams do not appear

covars Covariate matrix
traits Traits of interest, matrix
blocks Study design blocks, matrix
prior List of prior hyper-parameters

Details

You can use this data format for one-generation study designs, with option `alt=TRUE` in `MH`.

In these data, the component `ped` is a pedigree matrix with five columns: ID, sire, dam, sire's population and dam's population. This way you avoid typing extra rows for F generation, and it is also slightly faster to run than `specimen$ped`. The order of individuals should be the same in all data matrices (`ped`, `covars`, `traits` and `blocks`). Same individuals are used in `specimen` and `specimen2`, but there's more information in `specimen`, because it contains observations from F generation.

The component `prior` consists of lists. First component is the mean and variance of the fixed and population effects. You can replace the covariance matrix by `NA` for a theoretically justified model. The components `$G` and `$E` give the degrees of freedom and the inverse scale matrix of the respective priors.

Component `$random` is a list such that one component is a similar parameter list as is used for `G` and `E` matrices. In this case, `length(specimen$prior$random)` is 1, because there is only one random effect, that of design blocks. If you use more columns (and custom priors), you need a longer list. If you use your own priors, it is your responsibility to check that they are conformable with the study design.

Examples

```
# How it looks:
data(specimen2)
head(specimen2$ped)
head(specimen2$traits)
n = nrow(specimen2$ped)

# Running MH:
A <- specimen2
samp <- MH(A$poster, A$ped, A$covars, A$traits, 21, 7, 2, alt=TRUE)
```

threespines

Data from threespine sticklebacks

Description

Threespine sticklebacks from Lake Pulmanki, Lake Vattern and Baltic Sea.

Usage

```
data(threespines)
```

Format

List of six:

poster RAJM output, dimension: population x population x iteration

ped Pedigree matrix as in specimen2

covars Covariate matrix, only IDs

traits Four morphological traits

blocks Study design blocks, matrix

prior Empty list, amounts to default priors

Details

These data were used as an example by Karhunen et al. (2013).

Source

Data from Ecological Genetics Research Unit, University of Helsinki.

References

Karhunen et al. (2013): driftsel: an R package for detecting signals of natural selection in quantitative traits. *Molecular Ecology Resources* 13: 746-754.

See also Makinen et al. (2006): Genetic relationships among marine and freshwater populations of the European three-spined stickleback (*Gasterosteus aculeatus*) revealed by microsatellites. *Molecular Ecology* 15: 1519-1534.

Examples

```
data(threespines)
hist(threespines$traits[,2])
```

viz.theta

Illustrating neutral genetic differentiation

Description

This function illustrates the posterior of coancestry matrix theta as a planar graph.

Usage

```
viz.theta(thetapost, distance = T, center = F, main = NA)
```

Arguments

thetapost	posterior of theta
distance	one of TRUE, FALSE or NA, see below
center	plot ancestral mean at origin?
main	overall title, optional

Details

This function calculates the expected (absolute value of) difference in population means for each pair of populations under random genetic drift. Measuring the distance in ancestral standard deviation implies that this depends only on the coancestry matrix. This calculation produces a distance matrix. We use multidimensional scaling to present the distance matrix in a plane.

Using distance=T types the expected differences on the edges of the graph, while distance=F types the coancestry coefficients which depend inversely on the distances. distance=NA omits the edges.

You can use center=T to increase readability of the graph. This puts the ancestral population to origin, even though this is not in line with MDS as such.

Note that you can use viz.theta for both the posterior of theta obtained from RAFM and one obtained from driftsel::MH. The graph may change, because the quantitative traits also convey some information regarding the pattern of coancestry.

Value

Does not return anything.

Examples

```
data(specimen)
A <- specimen
samp <- MH(A$poster, A$ped, A$covars, A$traits, 21, 7, 2)
viz.theta(A$poster) # based only on molecular markers
viz.theta(samp$theta) # based on molecular markers & phenotypes
```

viz.theta.exact	<i>Illustrating neutral genetic differentiation</i>
-----------------	---

Description

This function illustrates the posterior of coancestry matrix theta as a planar graph. Unlike viz.theta, this produces the exact coordinates of the vertices.

Usage

```
viz.theta.exact(thetapost, distance = T, center = F, main = NA)
```

Arguments

thetapost	posterior of theta
distance	one of TRUE, FALSE or NA, see below
center	plot ancestral mean at origin?
main	overall title, optional

Details

For detailed documentation, see viz.theta. The only difference between these two functions is that viz.theta uses an approximation which nullifies the computation time. (It takes posterior mean of coancestry matrix prior to multidimensional scaling. viz.theta.exact does multidimensional scaling for every iteration of the coancestry matrix, and calculates the posterior mean of the coordinates.)

Value

Does not return anything.

viz.traits	<i>Population means in trait-trait space</i>
------------	--

Description

This is for visualizing the pattern of phenotypic differentiation.

Usage

```
viz.traits(fixedpost, popefpost, Gpost, THpost, traits, siz=0.5, main=NA)
```

Arguments

fixedpost	posterior of fixed effects from function MH
popefpost	posterior of population effects
Gpost	posterior of G matrix
THpost	posterior of theta
traits	which two traits to plot, integer vector
siz	probability mass within ellipses
main	an overall title

Details

The numbers represent population-level additive genetic effects, 'population means'. Each ellipse represents the median drift distance for the population of same color. (The ellipses have different sizes, because the local populations have different intra-population thetas, and thus experience different amounts of genetic random drift.) The ellipses are centered around the ancestral mean of the trait pair. All parameters refer to posterior means; estimation uncertainty is not presented in this figure.

In past, the probability mass within ellipses has caused some confusion. siz=0.5 produces the classic driftsel image where the ellipse is merely the borderline between 'stabilizing' and 'diversifying' and does not allow for hypothesis testing. siz=0.95 would show the 0.95 probability set under random genetic drift. siz=0.05 produces the region that is particularly indicative of stabilizing selection.

Value

Does not return anything.

Examples

```
data(specimen)
A <- specimen
samp <- MH(A$poster, A$ped, A$covars, A$traits, 21, 7, 2)
viz.traits(samp$fixed.ef, samp$pop.ef, samp$G, samp$theta, 1:2)
viz.traits(samp$fixed.ef, samp$pop.ef, samp$G, samp$theta, 1:2, siz=0.95)
```

Index

*Topic **datasets**

- specimen, [7](#)
- specimen2, [8](#)
- threespines, [9](#)

driftsel, [2](#)

driftsel-package (*driftsel*), [2](#)

H.test, [3](#)

MH, [4](#)

neut.test, [6](#)

S.test, [6](#)

specimen, [7](#)

specimen2, [8](#)

threespines, [9](#)

viz.theta, [10](#)

viz.theta.exact, [11](#)

viz.traits, [12](#)